



Cross-border venture capital investments in Asia: Selection and exit performance[☆]

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ABSTRACT

We investigate the investment behavior and exit performance of VCs that have pursued expansion outside their home locations, specifically, in Asia. Our findings indicate that, in the Asian VC markets, foreign VCs have relative advantages over local VCs in terms of size and experience while they are at a disadvantage in information collection and monitoring due to both geographic and cultural distances. When investing alone, foreign VCs are more likely to invest in more information-transparent ventures. Partnership with local VCs helps alleviate information asymmetry and monitoring problem and has positive implication for the exit performance of local entrepreneurial firms. Specifically, we find that after controlling for the endogeneity of selection, firms with both foreign and local VC partnership are about 5% more likely to successfully exit.

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1. Executive summary

The internationalization of venture capital (VC) investments has become a trend recently. The emerging markets such as China and India, where the VC industries are very much underdeveloped, have attracted a lot of attention and interest from international VCs. Among the various risks and challenges in the cross-border venture capital investments, the information asymmetry and moral hazard that arise due to geographical and cultural distances seems to be a universal concern. There is a dearth of empirical work on how frictions associated with geographic distance and different cultural environment impact VCs' investment behavior and how VCs respond to these issues, as well as how effective these solutions are, with a few exceptions which focus on developed economies such as U.S. and European area (Guler and Guillen, 2010; Manigart, 1994; Manigart et al., 2002; Meuleman et al., 2009;

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Sapienza et al., 1996). Our paper fills this gap by empirically examining investments by foreign VCs and their performances in Asian countries during the 1996–2006 period. We empirically examine a number of hypotheses that relate to the formation of different venture ownership types and their impact on the exit performance of the entrepreneurial firms. Specifically, we consider the following factors, the information friction and monitoring cost associated with both the geographic and cultural distances of the VCs to the companies funded, and the partnership between foreign and local VCs.

Our empirical results show that in the Asian VC market, foreign VCs play an important role. More than 70% of the \$35 billion funding in the Asian VC market from 1996 to 2006 is provided by foreign VCs. Foreign VCs invest in more information-transparent ventures when they invest alone. By forming partnership with local VCs, foreign VCs invest more aggressively in early stage ventures and ventures in technology industries than those acting alone. We further show that the cultural distances between the origin country of the foreign VC and the country of the local VCs not only discourages the formation of partnership between the two, but also has a negative impact on the exit performance of ventures.

We find that partnership between foreign and local VCs helps alleviate the frictions associated with geographical and cultural distances. Specifically, entrepreneurial firms jointly invested by foreign and local VCs are about 5.0% more likely to successfully exit than those invested by foreign VCs alone and by local VCs alone. This finding is robust after we control for the endogeneity of selection that more experienced VCs attract the better entrepreneurial deal and thus have a better chance of success using the propensity score matching methodology. In general, the findings in this paper shed additional light on the investment selection behavior and exit performance of VCs when they invest across borders. In particular, the paper studies how VCs deal with information asymmetry and moral hazard due to both geographical and cultural distances when investing in Asia, which is far less explored in the existing international venture capital literature. While it is well documented in the theoretical literature that cultural distance plays an important role in international diversification, the empirical findings are largely mixed. Our evidences suggest it has a negative impact on both the formation of partnership between foreign and local VCs and the exit performance of ventures. Our study extends the syndication literature by showing that partnership produces synergy in cross-border venture capital investments by reducing frictions associated with both geographical and cultural distance. The study also has implications for practitioners interested in investing in the Asian VC market.

2. Introduction

Cross-border venture capital (VC) investments are increasing manifold in terms of number of deals and capital involved. Surveys on the global trends in venture capital investments show that more than half of U.S. VCs plan to expand internationally, particularly in China and India (Deloitte, 2006, 2007). This trend raises several practical challenges for both investors and entrepreneurs. Among the various risks and challenges, a lack of knowledge or expertise in the local business environment due to geographical and cultural distances seems to be a universal concern. The trend also raises several questions that are not thoroughly studied in the existing academic literature. For instance, how do frictions associated with geographic distance and different cultural environments impact VCs' investment behavior? Does syndication between foreign and local VCs help reduce these frictions? How successful are these cross-border investments in terms of exit performance?

There is a dearth of empirical work on these questions, with a few exceptions which focus on developed economies such as U.S. and European area (Guler and Guillen, 2010; Manigart, 1994; Manigart et al., 2002; Meuleman et al., 2009; Sapienza et al., 1996). Our paper addresses the above questions by empirically examining investments by foreign VCs in Asian countries during the 1996–2006 period. We focus our analysis on the Asian market for the following reasons. First, this market is young and relatively underdeveloped and thus relatively less studied in the literature with a few exceptions (Bruton and Ahlstrom, 2003; Bruton et al., 2005; Bruton and Lau, 2008; Wright and Lockett, 2003; Young et al., 2008). Second, given the undeveloped nature of the local venture capital industry and the uniqueness of cultural and social norms in Asia, it provides an excellent setting to examine how foreign (non-Asian) VCs respond to the substantial frictions associated with both geographic distances and cultural disparity, and the effectiveness of their responses. Third, given the fast growth and increasingly important economic power of China and India, this area has attracted and continues to attract many foreign investors including foreign venture capital investors. Our study thus provides timely insights for the practitioners planning such expansion in this area.

We examine a number of hypotheses that relate to the formation of different venture ownership types and their impact on the exit performance of the entrepreneurial firms. Specifically, we consider the following factors, the information friction and monitoring cost associated with both the geographic and cultural distances of the VCs to the companies funded, and the partnership between foreign and local VCs.

We start by examining the differences between foreign and local VCs in the Asian markets. Our empirical results indicate that in the Asian VC market, foreign VCs play an important role. Among the total of \$35 billion investment in the Asian VC market reported by VentureXpert, from 1996 to 2006, more than 70% of the funding is provided by foreign VCs.³ There are partnerships between foreign and local VCs either through syndication in the same round of financing or funding the same venture in different rounds. However, foreign VCs, more often, act alone (without local VCs) in making investments in the Asian VC market.

Second, we examine how the frictions associated with geographic distance and cultural disparity impact VCs' behavior when they explore their investment opportunities in unfamiliar territory. We find that foreign VCs are less likely to invest in seed or early stage firms or participate in the very first round, which is consistent with these VCs avoiding more informationally opaque

³ The data reported by VentureXpert may not represent the true population.

investments. Partnership with local VCs seems to help alleviate such frictions to some extent. By forming alliances with local VCs, foreign VCs invest more in early stage ventures and ventures in technology industries than those acting alone. Furthermore, we find the cultural distances between the origin country of the foreign VC and the country of the local VCs negatively influences the formation of partnership between the two at the round level.

Third, we examine the exit performance implications of the above mentioned factors. We show that at the firm level, entrepreneurial firms jointly invested by foreign and local VCs are significantly more likely to successfully exit than those invested by foreign VCs alone or by local VCs alone. This finding is robust after we control for the endogeneity of the matching between VCs and firms using the propensity score matching. Specifically, firms with both foreign and local VCs as investors are about 5% more likely to successfully exit. Our finding suggests that in international venture capital investments, partnership between foreign and local VCs can provide potential synergy in exit performance, consistent with the evidence based on U.S. domestic venture capital investments (Hochberg et al., 2007; Sorenson and Stuart, 2001).

In our view, by studying the investment strategy and exit performance when VCs invest across borders, we contribute to the literature in several ways. First, this paper deepens the understanding of VCs' investment behavior while they are pursuing or expanding their international investing focus, and thus adds to the research focusing on issues related to the internationalization of VCs (Bruton and Lau, 2008; Makela and Maula, 2006, 2008; Wright et al., 2005). We find that due to frictions associated with geographic and cultural distances, foreign VCs are more likely to invest in informationally transparent firms while they are investing alone. Foreign VCs make efforts to develop partnerships with local VCs to overcome the frictions associated with distance, which has a positive exit performance implication.⁴ Second, our paper contributes to the foreign direct investment literature. Most of the existing studies focus on the manufacturing sector (Aitken and Harrison, 1999; Javorcik, 2005; Petkova, 2008). The internationalization of the VC/Private Equity (PE) industry is distinctive from the manufacturing sector or other professional and financial service sectors (Meuleman and Wright, 2011 and Wright et al., 2005). In contrast to the aim of managerial control in the traditional foreign direct investment, VC firms' ultimate purpose is to realize the financial returns, and consequently, exit is strategically important for VCs. On the other hand, VCs assume an active role, mentoring, advising and monitoring the entrepreneurs, to add value to the entrepreneurial firms. Therefore, the information asymmetry and moral hazard associated with distance and cultural disparity is particularly acute. Our study provides distinct empirical evidence to this literature by investigating VCs' behavior when investing across borders. Third, our paper provides new evidence to the literature on cross-border VC/PE syndication (Makela and Maula, 2006, 2008; Manigart et al., 2006; Meuleman et al. 2009; Meuleman and Wright, 2011). We examine the effect of the VCs' organizational form, age, experience, and the geographic and cultural distance between the origin country of the foreign VC and the country of the local VC on venture selection and exit performance. We further extend the literature by showing that partnership with local investors is significantly related to the ultimate exit performance of the ventures. Fourth, our paper is also related to the development of the VC industry in the emerging Asian market. In particular, our paper adds unique empirical evidence regarding the VC industry in Asia which has been very much underdeveloped but at the same time is becoming increasingly attractive market (especially China and India) for international VCs. Our findings thus have implications for the practitioners regarding the strategy to invest in the Asian VC market.

The remainder of the paper is organized as follows. Section 3 provides a brief literature review and develops the hypotheses. Section 4 describes the sample and presents the summary statistics. In Section 5, we empirically examine the investment criteria of foreign VCs and the exit performance implications of VC ownership type. Finally, in Section 6, we summarize our findings, explore their implications, and discuss their limitations.

3. Literature review and hypotheses development

In this paper, we study the investment behavior of VCs while they pursue expansion outside their home locations, specifically, VCs expanding their investments to the emerging Asian markets. We examine a number of hypotheses that relate to the formation of different venture ownership types and their impact on the exit performance of the entrepreneurial firms, with a focus on the information friction and monitoring cost associated with both the geographic and cultural distances of the VCs to the companies funded.

High-technology entrepreneurship in emerging economies has greater resource constraints and more informality than in developed economies (Siqueira and Bruton, 2010). Similar constraints and informality might appear in the venture capital industry of the emerging economies. The venture capital industry is, in general, rather young and underdeveloped in the Asian markets. There are a small number of local VC firms, which are often operating on a small scale and not able to offer much value-adding advice to the entrepreneurial firms due to their limited experience (Bruton and Ahlstrom, 2003). Foreign VCs from U.S. and other developed economies, on the other hand, have relatively rich experience developed in their home countries and are more resourceful in terms of amount of capital and networks they possess. Albeit foreign VCs may have relative advantage in experience and resource, they are often constrained by the information friction due to the geographic distance (Deloitte, 2010).

⁴ In contemporaneous research, Chemmanur et al. (2010) examine the impact of geographic distance on venture performance when VCs invest internationally. Our study is distinct from theirs in several aspects in addition to the different geographic focus. For instance, their analysis focuses on the impact of geographic distance, whereas our study examines both geographic and cultural distance. Moreover, we examine how geographic and cultural distances impact both the selection and exit performance, while they focus on the performance implication.

There is a growing literature that suggests that distance matters for information-collection-based financial activities. For instance, in small business lending, geographic proximity remains an important determinant of how businesses and lenders match with one another even in the era with advanced information technology (Rajan, 1992; Peterson and Rajan, 2002). U.S. mutual fund managers prefer to own stocks of companies located nearby and earn higher risk-adjusted returns (Coval and Moskowitz, 1999; 2001). Local households earn significantly greater returns from local holdings relative to more distant holdings, suggesting that local investors can exploit local knowledge (Ivkovic and Weisbenner, 2005). Similar results are also found in the setting of financial analysts and block acquisitions (Kang and Kim, 2008; Malloy, 2005). All these evidences point to the importance of the geographic proximity in investments due to the information advantage and reduced monitoring cost.

In a similar vein, studies find that geographic distance matters in venture capital investments. For instance, ventures in the U.S. proximate to their VCs are more likely to exit successfully via IPOs or M&As when controlling for the quality of ventures and VC reputation (Cumming and Dai, 2010).

Foreign VCs – by virtue of headquarters location in another nation – may be less able to process “soft” information about opaque local firms and/or their local market conditions—and therefore would be more likely to use their advantages in processing “hard” information and enter into relationships with more transparent firms. The geographic distance also makes close monitoring difficult. To reduce agency cost and moral hazard, foreign VCs would again avoid investing in informationally opaque firms. On the other hand, local VCs have information advantages over foreign VCs. They may be more able to process “soft” information about local firms and have better knowledge of local market conditions. It is also less costly for them to monitor the local entrepreneurs. In summary, we expect:

H1. Foreign VCs are more likely to invest in information-transparent firms due to the frictions associated with geographic distance (for instance, they tend to invest in later financing rounds, later stage ventures, and ventures in non-technology industries).

It is well established that cultural distance plays an important role in international diversification. However, the findings are mixed. For instance, some studies show that the relation between cultural distance and international diversification was negative for high-technology industries but positive for all other industries (Tihanyi et al., 2005). Shenkar (2001) indicates that even in the same culture, entrepreneurs have different beliefs and values defined by psychic distance that moderates cultural distance and its effects in foreign location decisions. Related to this context, an explanation is offered to understand these conflicting results reached by different studies on the relationship between cultural distance and international diversification (Brouthers and Brouthers, 2001).

Cultural influences and social norms, nevertheless, are often neglected in understanding the behavior of VCs in studies on the internationalization of venture capital investments (Wright et al., 2005). In this regard, we examine how the cultural disparity between foreign VCs and local firms, as well as local VCs, might impact VCs' investment behavior and their exit performance.

Several studies show that cultural dimensions have important implications for the development of entrepreneurship. For instance, individualism and collectivism link to different functions of entrepreneurship (Tiessen, 1997). Specifically, individualism is primarily associated with variety generation that supports the diversity of opportunity recognition for local entrepreneurial firms, while collectivism contributes mostly to resource leverage indicating the quest by foreign capital for syndication possibilities. Cultural dimensions also have important implications for two critical attributes of entrepreneurship, internal locus of control and innovativeness (Mueller and Thomas, 2000). Gupta et al. (2004) propose that entrepreneurial leadership is related with Hofstede's (2001) cultural scales. Specifically, cultures characterized by high power distance are less likely to endorse entrepreneurial leadership. The authors suggest that when VCs invest in ventures from a different culture, they could be facing a very different group of entrepreneurs with distinctive personalities, skills, expectations, working styles, among other things, from the entrepreneurs in their home countries.

On the other hand, VCs from different cultures may have developed their own preferred investment philosophies and value-adding skills, which are presumably most fit to the entrepreneurial firms in their home nations. For instance, VCs from different cultures differ in their monitoring activities (Pruthi et al., 2003). Specifically, Indian domestic VCs regard the relationship between entrepreneurs and VCs as part of a unified network rather than an arms' length agency one due to the emphasis on shared responsibility in the culture. Thus monitoring is of less importance to Indian domestic VCs, in contrast to U.S. VCs. When VCs invest in a different culture, some of their investment styles and value-adding skills developed in their home countries could be universally useful, while others may not fit the new culture. Bruton et al. (2005) indicate that the U.S. VCs follow the model they developed at U.S. even when they operate in a different environment. The challenges encountered by VCs and the investment framework employed in China, however, is very different from the U.S. (Bruton and Ahlstrom, 2003). For instance, some of the VCs' monitoring activities, which are commonly employed in the U.S., are regarded as intrusive in China and receive much resistance. Thus, VCs' prompt adjustment to the local culture when they expand internationally could be critical for the success of their investments. Companies going overseas suffer lower returns until they become accustomed to the local culture and networks (Contractor et al., 2003).

We conjecture that the ability of VCs to quickly and effectively adapt their investment philosophies and value-adding skills to entrepreneurs or entrepreneurial firms from a different culture is likely to be conditional on the distance between the cultures of the VC's country and the entrepreneur's country. The greater the cultural distance, the greater difficulty and more time it takes for such adjustments. Before foreign VCs can successfully understand and adapt to the local culture, there likely would be serious distrust between foreign VCs and local entrepreneurs, as well as local VCs. For instance, studies show that countries with higher cultural distance display higher mutual distrust (Chakrabarti et al., 2008; Guiso et al., 2008). This distrust due to cultural disparity

would be an obstacle to the formation of partnership between foreign and local VCs. For instance, a VC from U.S. investing in China potentially will have relatively greater difficulty in understanding the Chinese cultural and social norms than a Singapore VC. The similarity in cultural background between a Singapore VC and a Chinese local VC potentially will facilitate forming a partnership between them and the effectiveness of such partnership. Further, various frictions may arise from the distrust between foreign VCs and local entrepreneurs, such as hiding key information from each other, misunderstanding due to ineffective communication, which would potentially mitigate the effectiveness of VCs' monitoring and other value-adding activities, thus hurting the exit performance of entrepreneurial firms.

Following the existing studies, we use Hofstede's (2001) measures of the four dimensions of a society's culture, and then use the differences in the measures to capture the idea of "cultural distance" between countries.⁵ We expect that cultural distance impacts VC partnership formation and the exit performance of the entrepreneurial firms in the following way.

H2a. The greater the cultural distance between the origin country of foreign VCs and the country of the local VCs, the less likely the foreign and local VCs will form partnerships.

H2b. The cultural distance between the origin country of foreign VCs and the country of the local entrepreneurial firms is inversely associated with ventures' exit performance.

According to the Deloitte's surveys, VCs that pursue international investing are aware of the additional risk and challenges associated with the physical distance and the cultural disparity. To overcome these risks, in practice, foreign VCs have developed several strategies. For example, some VC firms explore the investment opportunities in foreign countries by establishing a local branch office and hiring local professionals. For instance, Softbank China Venture Capital is a branch office of Softbank Venture Capital, which is headquartered in California. Some Foreign VCs form partnerships with local VC firms to reduce information asymmetry and co-manage local investments. For example, Mayfield, a major VC firm in U.S., is an investor in GSR Ventures, a local Chinese VC firm, and GSR Ventures manages Mayfield China (a China fund affiliated with Mayfield) together with Mayfield. A third approach is to syndicate with independent local VC investors in a specific investment. For instance, Walden International, headquartered in the U.S., and Ceyuan Ventures, a local VC firm in China, co-invested in 3GPP, a Chinese venture. Ideally, we would like to examine all three approaches and their effectiveness in reducing information asymmetry and agency cost in international VC investment. Given the lack of sufficient data, we focus our analysis on the third approach, the syndication between foreign and local VCs.⁶

Syndication between foreign and local VCs potentially will expand the pool of investment opportunity. VCs form syndication primarily for the purpose of better investment selection (Lerner, 1994). In U.S. venture capital investments, syndication between distant VCs and local VCs helps overcome information asymmetry and agency problems (Sorenson and Stuart, 2001). In the setting of international VC investments, we expect that foreign VCs could benefit from forming partnerships with local VCs in terms of reducing information asymmetry, improving effectiveness of monitoring, and thus expanding their investment selection pool. Specifically, we expect that the partnership between foreign and local VCs will make it possible to invest in more informationally opaque local ventures than if foreign VCs invest alone.

Partnership between local and foreign VC firms can provide potential synergy in exit performance as well. Many studies document a positive impact of syndication on the ventures' performance arguing that VC syndication provides a wide range of skills and networks to portfolio companies (Brander et al., 2002; Das et al., 2011; Gompers et al., 2009; Hochberg et al., 2007; Wright and Lockett, 2003). In international venture capital investments, firms with both foreign and local VCs on board will not only share all the benefits of syndication mentioned above, but also benefit from the unique structure of syndication in at least the following three aspects. First, the broad experience and skills, the combined knowledge of both the local business environment and the international capital market, and additional resource from foreign and local VCs partnership will amplify VCs' value addition. Second, the reduced frictions due to geographic and cultural distances with the assistance from local VCs allow better selection and monitoring. Third, for the purpose of risk sharing, when investing in an uncertain investment environment, joint ventures may also involve a greater number of investors. Larger syndicate size further suggests positive implication for the exit performance as shown in many previous VC networking and syndication studies (Brander et al., 2002; Das et al., 2011; Gompers et al., 2009; Hellmann and Puri, 2002; Hochberg et al., 2007; Lerner, 1994; Lindsey, 2008). In summary, we expect the following.

H3a. Forming partnership with local VCs may allow foreign VCs to reduce information friction more effectively and thus, they are willing to invest in more informationally opaque local firms.

H3b. Firms invested jointly by foreign and local VCs may have better exit performance than those invested by local VCs or foreign VCs alone.

⁵ The mathematical formula for estimating cultural distance is introduced in the empirical section.

⁶ Bouquet et al. (2004) suggest that international companies gravitate towards wholly owned subsidiaries and expatriate managers if close relationships with end customers and a high level of skills are necessary for successful operations. Anecdotal evidence shows that many VC companies either expatriate their managers or hire local talents to make investment and monitoring smoother. However, these data are not collected by the major venture capital databases including VentureXpert. A future research potentially using hand-collected data would help fill this gap.

Table 1

Venture capital investments in Asia, 1996–2006.

In this table, we summarize the amount of capital invested by foreign and local VCs in six Asian economies, including China, Hong Kong, Taiwan, India, Singapore, and South Korea. We also present the number of rounds and the number of ventures that are financed by foreign VCs alone, local VCs alone, and jointly by foreign and local VCs. The mean and median round sizes are also reported.

	China	Hong Kong	Taiwan	India	Singapore	South Korea	Total
Total capital (in \$B)	9.1	3.1	1.7	7.1	4.6	9.8	35.4
Foreign (in \$B)	8.1	2.2	1.5	5.4	3	6.2	26.4
Local (in \$B)	1	0.9	0.2	1.7	1.6	3.6	9
Number of rounds	581	202	173	928	266	2104	4254
Foreign	384	116	87	265	163	145	1160
Local	68	49	47	534	56	1629	2383
Joint	129	37	39	129	47	330	711
Number of ventures	418	135	134	640	169	1364	2860
Foreign	249	68	63	180	86	69	715
Local	60	36	35	332	42	1013	1518
Joint	109	31	36	128	41	282	627
Mean round size (\$M)	15.5	15.1	9.8	7.6	17.3	4.6	8.3
Median round size (\$M)	5.5	5	2.6	1.5	2.9	0.6	1.1

4. Data and measurement

Our sample includes 4254 rounds of venture capital financing by 468 venture firms in the following Asian countries or districts: Mainland China (China hereafter),⁷ Hong Kong, Taiwan, India, Singapore, and South Korea from 1996 to 2006 as documented by Thomson Financial's Venture Economics (VentureXpert) Database.⁸ The IPO and acquisition information is also collected from Thomson Financial's SDC Platinum Mergers and Acquisitions (M&As) and Global New Issues Database.

As shown in Table 1, a total of 2860 venture companies received venture capital financing, including 640 ventures located in India, and 418 ventures in China. Over the 11-year period, a total of \$35.4 billion of venture capital has been invested in this area that includes \$26.4 billion invested by foreign VCs. The average round size ranges from \$4.6 million (South Korea) to \$17.3 million (Singapore) and the median round size ranges from \$0.6 million (South Korea) to \$5.5 million (China).

Most of the foreign VCs are US VCs. In our untabulated results, a total of 247 out of 468 foreign VCs (52.8%) are US VCs. They have participated in 1586 rounds, with a total invested amount of \$17.9 billion, which is 67.8% of all foreign venture capital invested in these markets. Asian VCs' cross investment is also prevalent. A total of 147 foreign VCs are from the nearby Asian countries. This group of VCs has participated in 501 rounds of new ventures and has invested \$6.6 billion in total. The third group of foreign VCs is from various countries other than US and Asia. This group invested in 242 ventures a total of \$1.9 billion.

We define venture ownership type at two levels. First, at the round level, *foreign VC investment* refers to financing rounds that are invested by foreign VCs alone; *local VC investment* refers to financing rounds that are invested by local VCs alone; *joint investment* represents those financing rounds that are invested by foreign and local VCs together. Second, at the entrepreneurial firm level, *foreign VC investment* refers to ventures that are invested by foreign VCs alone in all rounds; *local VC investment* refers to ventures that are invested by local VCs alone in all rounds; *joint investment* represents those ventures with both foreign and local VCs investing in the venture.

In Table 1, we categorize VC financing rounds and entrepreneurial firms based on VC ownership type. About 27% of the 4254 financing rounds are invested by foreign VCs alone, about 56% are invested by local VCs alone, and the remaining investments are by partnerships of foreign and local VCs. Some countries seem to rely on foreign VCs for funding of young private firms more heavily than local VCs. For example, in China, foreign VCs participate in 88% of the financing rounds, among which 66% are financed by foreign VCs only, and only 12% of the financing rounds are purely financed by local VCs. Among the 418 China local ventures, 86% have certain percentage of foreign VC ownership, among which 60% are alone financed by foreign VCs, while only 14% are alone financed by local VCs. On the other hand, in South Korea, 74% of the young private firms are purely invested by local VCs, while only 5% of the entrepreneurial firms are alone invested by foreign VCs.⁹

⁷ Here, Mainland China refers to the People's Republic of China excluding Hong Kong and Macau. We use China and Mainland China interchangeably in the paper.

⁸ As VentureXpert data are somewhat unreliable and less complete in the early period, we focus our analysis on the relatively recent period. We further discuss the potential biases of the database and their implications for our findings in greater details in Section 6.

⁹ Mainly due to government policy, detailed regulation, and relatively smaller size of entrepreneurial market in Korea during the sample period of 1996–2006, there are so few foreign VC deals in South Korea. The main reasons of the smaller size of Korean entrepreneurial market are that they are on average younger and less successful than U.S. ones, but another reason is that large asset management institutions such as pension funds and insurance companies have not participated yet in the VC partnerships in Korea (Jo et al., 2008). Compared to other countries, South Korea is clearly an outlier. Thus, we have conducted robustness checking by excluding South Korea from the sample, and obtain qualitatively similar results.

5. Empirical analysis

5.1. Univariate analysis

Table 2 presents the univariate analysis of venture characteristics associated with different VC ownership type. We use three groups of measures as proxies for the level of information asymmetry and uncertainty. The first measure is the round number. The first round of financing typically has greater risk. With the certification of first round VC investors, later rounds are relatively less risky (Lerner, 1994). The second measure is the life stage of ventures at the time they receive venture capital investment. We define the following four stages: startup/seed stage, early stage, expansion stage, and later stage. Ventures at the early stages have higher level of information asymmetry and uncertainty than ventures at the later stages. The third measure is industry. Ventures in high-technology industries have greater level of information asymmetry and uncertainty. We identify the following six industry groups: computer-related, communication and media, semiconductor, biotechnology, medical/health care, and non-technology.

Table 2

VC ownership type and venture characteristics.

In this table, we relate venture characteristics to ownership type at both the round level and at the firm level. Specifically, we examine round number, stage, and industry of the local entrepreneurial firms. In Panel C, we also summarize the average round size and the total number of rounds firms received conditional on VC ownership type. *p*-values of F-tests for cross-group differences in means are provided in the last column. ***, **, and * denote statistical significance at 1%, 5%, and 10% confidence levels, respectively.

	Foreign		Local		Joint		F-test p-value
	N	%	N	%	N	%	
<i>Panel A: round level analysis</i>							
Round							
Round 1	721	62%	1802	76%	496	70%	0.000***
Round 2	231	20%	400	17%	138	19%	0.036**
Round 3 and plus	208	18%	181	8%	77	11%	0.000***
Stage							
Startup/seed	87	8%	326	14%	79	11%	0.000***
Early stage	226	19%	599	25%	211	30%	0.000***
Expansion	619	53%	1280	54%	324	46%	0.000***
Later stage	228	20%	178	7%	97	14%	0.000***
Industry							
Computer related	362	31%	704	30%	246	35%	0.001***
Communication/media	218	19%	425	18%	152	21%	0.015**
Semiconductor	119	10%	373	16%	113	16%	0.000***
Biotechnology	7	1%	106	4%	41	6%	0.000***
Medical/health care	47	4%	93	4%	22	3%	0.361
Non-technology	407	35%	682	29%	137	19%	0.000***
<i>Panel B: company level analysis</i>							
Stage when receiving first round financing							
Startup/seed	62	9%	249	16%	101	16%	0.000***
Early stage	148	21%	430	28%	210	33%	0.000***
Expansion	384	54%	740	49%	252	40%	0.000***
Later Stage	121	17%	99	7%	64	10%	0.000***
Industry							
Computer related	227	32%	479	32%	237	38%	0.000***
Communication/media	116	16%	269	18%	136	22%	0.000***
Semiconductor	68	10%	217	14%	90	14%	0.002***
Biotechnology	4	1%	69	5%	32	5%	0.000***
Medical/health care	26	4%	59	4%	19	3%	0.410
Non-technology	274	38%	425	28%	113	18%	0.000***
Syndicate							
1 investor	239	62%	1024	82%	0	0%	0.000***
2 investors	72	19%	163	13%	171	36%	0.000***
≥3 investors	72	19%	66	5%	303	64%	0.000***
<i>Panel C: investment size—company level analysis</i>							
	Foreign		Local		Joint		F-test p-value
	Mean	Median	Mean	Median	Mean	Median	
N of rounds company received	1.6	1.0	1.3	1.0	1.9	1.0	0.000***
Average round amount (\$M)	16.6	5.3	2.4	0.6	11.4	2.9	0.000***
Total amount invested in company (\$M)	25.4	7.5	2.9	0.7	22.6	4.4	0.000***

Panel A of Table 2 suggests that foreign VCs allocate smaller percentage of their investments to first round financing. For instance, 76% of local VCs' investments are first round financing, while only 62% of foreign VCs' investments are. We also find that foreign VCs are less likely to invest in seed and early stage ventures. For example, about 39% of local VCs' investments are allocated to seed and early stage ventures, while only 27% of foreign VCs' investments are for ventures at seed or early stages. Furthermore, foreign VCs are more likely to invest in non-technology ventures. About 35% of foreign VC investments are allocated to ventures in non-technology industries, while only 29% of local VC investments are made to non-technology ventures. Interestingly, this percentage is even lower (19%) when foreign and local VCs form partnership. At the company level reported in Panel B of Table 2, we find similar patterns. In comparison to ventures invested by local VCs alone, ventures with investments from foreign VCs only, often receive their first round of financing at their expansion stage or later stage. They are also more likely to be non-technology ventures. However, when foreign VCs invest jointly with local ones, we see significantly higher percentage of seed and early stage ventures (49%) and technology ventures (82%) in their portfolios, higher than both foreign VCs only and local VCs only. Furthermore, we show that joint investments are associated with more investors or larger syndicate size. Specifically, about 64% of the joint investments have more than or equal to three investors, while this percentage is 5% for the ventures invested by domestic VCs alone, and 19% for the ventures invested by foreign VCs alone.

In Table 2 Panel C, we compare the investment size by foreign VCs and local VCs. We find that ventures with foreign VC participation, in general, receive more rounds of financing. Average round size is also greater than those invested by local VCs. For instance, the mean and median round sizes for ventures financed by foreign VCs alone are \$16.6 million and \$5.3 million respectively, while they are \$2.4 million and \$0.6 million for ventures invested by local VCs alone. Ventures invested by foreign and local VCs jointly have investment size that is between these two values. This finding indicates that ventures with investments from foreign VCs overall have better accesses to financial capital.

Table 3 reports the frequency distribution of IPO and acquisition exits based on VC ownership type. Following the convention used by other researchers, we view a successful exit as a representation of the venture's success (Brander et al., 2002; Das et al., 2011; Gompers and Lerner, 1998, 2000; Hochberg et al., 2007; Sorenson, 2007). Specifically, we use data on IPOs and acquisition exits from 1996 to the end of 2007. Because it typically takes 5–6 years for VCs to exit from their investments, to avoid the potential bias, we exclude ventures that obtained their first round of financing after the end of 2002 for the purpose of analysis in this section. This leaves us 2101 ventures. Overall, about 10.4% (48 out of 461) of ventures financed by foreign VCs and about 11.0% (54 out of 490) of ventures jointly invested by foreign and local VCs went public successfully. In contrast, only about 7.1% (102 out of 1439) of ventures financed by local VCs alone went public. Ventures with foreign VC investments are also more likely to cross-list in foreign capital market. In fact, we do not find a single case where ventures financed by local VCs alone cross-list in foreign IPO markets. In terms of acquisition exits, we show that about 8.2% of ventures financed by foreign VCs alone and 5.1% of ventures jointly invested by foreign and local VCs were acquired by the end of 2007. In contrast, only about 2.6% of ventures financed by local VCs alone were acquired.

5.2. Multivariate analysis

When foreign VCs decide to invest in the international market, we view that there are actually two steps. First, VCs will decide whether they will enter a specific market. Then, they determine, in the second step, what would be the most effective ownership

Table 3

VC ownership type and ventures' ultimate exits (firm level).

In this table, we examine whether ventures had exited via an IPO or an acquisition by the end of 2007. Because it typically takes 5–6 years for VCs to exit from their investments, to avoid the potential bias, we exclude ventures that obtained their first round of financing after the end of 2002. This leaves us 2101 ventures.

	N of ventures	N of IPOs	%	N of foreign IPOs	%	N of acquisitions	%
Foreign VCs only							
China	87	9	10%	5	56%	4	5%
Hong Kong	71	7	10%	2	29%	2	3%
Taiwan	65	14	22%	1	7%	3	7%
India	89	8	9%	1	13%	12	14%
Singapore	89	5	6%	1	20%	12	14%
South Korea	60	5	8%	1	20%	5	8%
Local VCs only							
China	52	1	2%	0	0%	2	4%
Hong Kong	47	4	9%	0	0%	4	9%
Taiwan	51	14	27%	0	0%	0	0%
India	395	47	12%	0	0%	20	5%
Singapore	35	2	6%	0	0%	1	3%
South Korea	859	34	4%	0	0%	10	1%
Joint investments							
China	36	8	22%	7	88%	2	6%
Hong Kong	34	3	9%	2	67%	6	18%
Taiwan	26	6	23%	0	0%	2	8%
India	106	20	19%	1	5%	6	6%
Singapore	37	1	3%	0	0%	6	16%
South Korea	251	16	6%	1	6%	3	1%

type for a specific market. In the analysis that follows, we first analyze the choice of foreign VC participation (including both foreign VCs alone and joint ventures by foreign and local VCs) in comparison to domestic VCs alone and its implication for the ventures' exit performance. Then among the ventures with foreign VC participation, we further compare those that are jointly invested by foreign and local VCs with those that are invested by foreign VCs alone.¹⁰

5.2.1. Domestic VCs alone vs. foreign VC participation

In this section, we analyze the choice of domestic VCs alone and foreign VC participation (including both foreign VCs alone and joint investments) and its implication for ventures' exit performance. Panel A of Table 4 reports the results of probit regressions on the choice of domestic VCs alone and foreign VC participation at the round level and the firm level, respectively. In the first regression, the dependent variable is an indicator variable which is equal to 1 if a specific financing round has foreign VC participation, which could be foreign VCs alone or a syndicate of foreign and local VCs; and 0 otherwise. The independent variables include measures of information asymmetry: (a) round size and number; (b) the stage of ventures at the financing round date; (c) the industries in which the ventures reside. In addition, we control for lead VC characteristics such as independent VC dummy, Corporate VC dummy, whether the VC fund is a seed or early stage fund, VC age, and VC's total investment experience. The definitions of these variables are provided in details in the Appendix.

The dependent variable in the second regression is an indicator variable which is equal to 1 if at least one foreign VC participated in any of the ventures' financing rounds; and 0 otherwise. The independent variables are similar to those used in the round level analysis except that the lead VC is defined as the one that invests the largest amount in the very first round and stage dummies indicate the stage of the venture when they receive the very first round.

Further, in both specifications, we also control for country legality and stock market development. Legality is shown to be an important prerequisite for sustained VC development in a country and it impacts performance of entrepreneurial firms in emerging economies (Cumming et al., 2006; Cumming et al., 2010). The U.S. evidence shows that a well developed stock market is extremely important to the development of a venture capital industry as it provides a viable exit mechanism for both investors and entrepreneurs (Black and Gilson, 1998). Cross-country evidence shows that the level of stock market development matters for venture capital investment (Hazarika et al., 2010; Jeng and Wells, 2000). We measure the legality of a specific country as the weighted average of the following factors (based on Berkowitz et al., 2003, Cumming et al., 2006; Cumming et al., 2010): civil versus common law systems, efficiency of judicial system, rule of law, corruption, risk of expropriation, risk of contract repudiation and shareholder rights (as in La Porta et al., 1997; 1998; 1999). A higher value of legality index indicates a "better" legal system. We measure stock market development as the average of the three normalized indices, including stock market capitalization/GDP, total value of shares traded/GDP, and total value of shares traded/stock market capitalization (Khurana et al., 2006; Love, 2007; Wurgler, 2000). Finally, we include year and entrepreneurial firm country fixed effects in both specifications.

As shown in Panel A of Table 4, we find that at the round level, foreign VCs are less likely to invest in the first round and seed/early stage ventures. Specifically, ventures raising their very first round are 5% less likely to obtain foreign VC participation. The seed stage ventures and early stage ventures are about 13% and 6% less likely to obtain foreign VC investment, respectively. Furthermore, foreign VCs are more likely to invest in ventures with greater capital demand and computer-related ventures, as well as ventures in the communication and media industry. The firm level analysis (the second regression) provides results that are consistent with the round level analysis. For instance, we find that foreign VCs are 16–27% less interested in seed or early stage investments, 6% more likely to finance computer-related ventures and 10% more likely to invest in communication/media ventures. These findings are consistent with the univariate analysis and lend support to the idea that VCs are likely to behave in a way that is consistent with Hypothesis 1 where information asymmetry associated with distance reduces the possibility that foreign VCs invest in more information-opaque companies.

Panel A of Table 4 also provides further insight regarding the differences between foreign and local VCs. For instance, we show that the foreign VCs are often independent VCs or CVCs, while many local VCs are sponsored by governments or affiliated with other financial intermediaries such as banks. Foreign VCs investing in Asia are less likely to be a seed/early stage fund. In addition, we show that foreign VCs investing in Asia in general are more senior and more experienced than local VCs.

We also find that the better legality of the country where the entrepreneurial firms are located significantly and positively increases foreign VCs' investments in the local market, indicating that more solid legality helps local entrepreneurial firms attract foreign VC investment. Stock market development also significantly increases the probability of attracting foreign VCs' investments, consistent with the previous finding that U.S. VCs are more likely to invest in countries with developed financial institutions (Guler and Guillen, 2010).

In Panel B of Table 4, we compare the exit performance of ventures that are invested by domestic VCs alone and those with foreign VC participation in the multivariate setting. We analyze successful exits in three scenarios: IPOs only, IPOs and acquisitions, and IPOs and acquisitions with valuation data. While acquisition is a very important exit mechanism for both VCs and entrepreneurial firms, not all sell outs can be deemed as successful (for instance, fire sales). As acquisitions with larger transaction values are typically more likely to report valuation data, we use acquisitions with valuation data from the SDC database as a second-best proxy for successful acquisitions (at least, different from the deep-discount sales).¹¹

We start with the round level analysis, including the first three specifications in Panel B of Table 4. In the first regression, we apply the competing risk model, where the failure event is IPO exit and the competing event is unsuccessful exit. Those that have not exited are treated as right censored. The dependent variable of the competing risk regression is the natural logarithm of time to

¹⁰ In unreported analysis, we also apply multinomial logit framework to compare the three choices (domestic VCs alone, foreign VCs alone, and joint investments) directly. The empirical findings are in general similar. This set of results is available upon request.

¹¹ We would like to thank an anonymous reviewer for recommending this.

Table 4

Investments with foreign VCs vs. investments with domestic VCs alone.

In Panel A, we use probit regressions to investigate whether there is systematic difference in investment criteria across VC ownership types (investments with foreign VCs vs. investments with domestic VCs alone) at the round level and the firm level, respectively. The dependent variable in the first (second) regression is a dummy which is set to equal to one if at least one foreign VC invests in a specific financing round (in any of the financing rounds) and 0 otherwise. All regressions include entrepreneurial firm country fixed effect and year fixed effect. The independent variables include measures of information asymmetry: (a) round size and number; (b) the stage of ventures at the financing round date; (c) the industries in which the ventures reside. In addition, we control for lead VC characteristics such as independent VC, Corporate VC, whether the VC fund is a seed or early stage fund, VC age, and total investment experience. Definitions of other independent variables are provided in the Appendix. The independent variables in the second regression are similar to those used in the round level analysis except that the lead VC is defined as the one that invests the largest amount in the very first round and stage dummies indicate the stage of the venture when they receive the very first round. Marginal effects instead of coefficients are reported. In Panel B, we analyze the impact of VC ownership type (investments with foreign VCs vs. investments with domestic VCs alone) on the likelihood of successful exits. For successful exits, we consider three scenarios: IPOs only, IPOs and M&As, and IPOs and M&As with valuation data. The first three regressions represent the round level analyses. In the first regression, we apply the competing risk model, where the failure event is IPO exit and the competing event is unsuccessful exit. Those that have not exited are treated as right censored. In the second and third regressions, we utilize Cox Proportional Hazard model. The dependent variable of these three regressions is the natural logarithm of time to an exit which is measured from the date of each round. Hazard ratios instead of coefficients are reported. The independent variable of our key interest is the ownership type of the venture. Specifically, *foreign VC participation* is equal to 1 if at least one foreign VC invests in a financing round and 0 otherwise. Other independent variables that may have an impact on venture exit performance are also included, for instance, dummies indicating the stage of investment that includes seed, early and expansion; the type of industry with computer, communication, semiconductor, biotechnology and medical/healthcare as the choices; the lead VC characteristics with independent VC, corporate VC, VC age, VC's total investment experience, VC's investment experience in the entrepreneur's country, measures of geographic distance and cultural distance between VCs and entrepreneurs. The last three regressions are the firm level analyses. We utilize probit regressions in all three specifications, where the dependent variable is a dummy which is equal to 1 if the venture has exited as an IPO (IPO or Acquisition, IPO or Acquisition with valuation data) by the end of 2007, and 0 otherwise. At the firm level analyses, *foreign VC participation* is equal to 1 if at least one foreign VC invested in the venture in any of the rounds, and 0 otherwise. The independent variables are similar to those used in the round level analysis except that the lead VC is defined as the one that invests the largest amount in the very first round and stage dummies indicate the stage of the venture when they receive the very first round. All regressions include year fixed effect and entrepreneurial firm nation fixed effect. ***, **, and * represent significance at the 1%, 5%, and 10% confidence levels, respectively.

Panel A: selection		
	Round level	Firm level
Round		
Round size	0.253*** (0.000)	
Round 1	−0.045* (0.076)	
Stage		
Startup/seed	−0.126*** (0.007)	−0.274*** (0.000)
Early stage	−0.058 (0.160)	−0.161*** (0.000)
Expansion stage	−0.071* (0.053)	−0.172*** (0.000)
Industry		
Computer related	0.073** (0.015)	0.063** (0.048)
Communication and media	0.061* (0.078)	0.101*** (0.006)
Semiconductor	0.003 (0.937)	0.014 (0.738)
Biotechnology	−0.002 (0.974)	0.003 (0.969)
Medical/health care	−0.023 (0.694)	−0.054 (0.389)
Lead VC characteristics		
Independent VC	0.345*** (0.000)	0.319*** (0.000)
CVC	0.251*** (0.000)	0.215*** (0.000)
Seed or early stage fund	−0.066** (0.010)	−0.108*** (0.000)
Ln(VC firm age)	0.077*** (0.000)	0.114*** (0.000)
Ln(total investment experience)	0.121*** (0.000)	0.073*** (0.000)
Other control variables		
Country legality index	0.121*** (0.000)	0.112*** (0.000)
Stock market development	0.295*** (0.000)	0.172*** (0.000)
Year fixed effect	Yes	Yes
Entrepreneurial firm nation fixed effect	Yes	Yes
N	3876	1955
Pseudo R ²	54.00%	35.82%

(continued on next page)

Table 4 (continued)

Panel B: exit performance						
	Round level			Firm level		
	IPO	IPO and ACQ	IPO and ACQ (with valuation)	IPO	IPO and ACQ	IPO and ACQ (with valuation)
Ownership type						
Foreign Participation	1.410 (0.162)	1.831*** (0.009)	1.768** (0.015)	0.014 (0.409)	0.008 (0.699)	0.017 (0.369)
Stage						
Startup/seed	0.192*** (0.000)	0.412*** (0.002)	0.362*** (0.001)	−0.051*** (0.000)	−0.054*** (0.003)	−0.051*** (0.000)
Early stage	0.662* (0.099)	0.643** (0.035)	0.616** (0.025)	−0.031** (0.023)	−0.019 (0.403)	−0.036** (0.029)
Expansion stage	1.059 (0.784)	0.942 (0.739)	0.961 (0.824)	−0.024 (0.112)	−0.018 (0.432)	−0.023 (0.198)
Industry						
Computer related	1.193 (0.331)	1.132 (0.427)	1.064 (0.700)	0.008 (0.568)	0.018 (0.332)	0.013 (0.427)
Communication and media	0.795 (0.315)	1.117 (0.549)	1.061 (0.752)	0.004 (0.782)	0.033 (0.154)	0.022 (0.247)
Semiconductor	1.260 (0.311)	1.430* (0.075)	1.306 (0.193)	0.047* (0.056)	0.080** (0.016)	0.052* (0.058)
Biotechnology	1.133 (0.777)	0.695 (0.347)	0.704 (0.419)	−0.005 (0.853)	0.025 (0.571)	−0.010 (0.746)
Medical/health care	1.800* (0.061)	1.850** (0.036)	1.769* (0.052)	0.080* (0.072)	0.097* (0.064)	0.090* (0.064)
Lead VC characteristics						
Independent VC	0.791 (0.177)	0.808 (0.166)	0.813 (0.192)	−0.004 (0.743)	0.001 (0.998)	−0.003 (0.843)
CVC	0.871 (0.567)	0.852 (0.442)	0.872 (0.516)	−0.017 (0.192)	−0.011 (0.566)	−0.011 (0.490)
Ln(VC firm age)	0.905 (0.318)	0.953 (0.570)	0.956 (0.606)	0.001 (0.864)	0.007 (0.365)	0.009 (0.173)
Ln(total investment experience)	1.060 (0.347)	1.015 (0.776)	1.042 (0.432)	0.002 (0.546)	−0.001 (0.785)	−0.001 (0.896)
Ln(experience in entrepreneur's country)	1.001 (0.994)	1.031 (0.725)	0.993 (0.942)	0.003 (0.612)	−0.004 (0.671)	0.001 (0.921)
Ln(geographic distance)	1.251** (0.046)	1.119 (0.166)	1.117 (0.181)	0.003 (0.719)	0.007 (0.411)	0.001 (0.933)
Cultural distance	0.944*** (0.004)	0.959*** (0.008)	0.957*** (0.006)	−0.002 (0.266)	−0.001 (0.789)	−0.001 (0.435)
Other control variables						
Country legality index	1.026 (0.721)	1.070 (0.188)	1.074 (0.181)	−0.006 (0.182)	0.010** (0.046)	−0.002 (0.714)
Stock market development	1.336 (0.395)	1.054 (0.858)	1.141 (0.661)	0.050** (0.020)	0.031 (0.281)	0.043* (0.075)
Ln (total investment)	1.333*** (0.000)	1.282*** (0.000)	1.342*** (0.000)	0.019*** (0.000)	0.035*** (0.000)	0.028*** (0.000)
Year fixed effect	Yes	Yes	Yes	Yes	Yes	Yes
Entrepreneurial firm nation fixed effect	Yes	Yes	Yes	Yes	Yes	Yes
N	3876	3876	3876	1955	1955	1955
Pseudo R ²				11.75%	11.68%	11.40%
Log pseudolikelihood	−1753.05	−2121.24	−2023.22			
Wald Chi-square	247.36	233.76	240.84			
Prob>Chi-square	0.000	0.000	0.000			

an exit which is measured from the date of each round. We utilize the Cox Proportional Hazard model in the second and third regressions. The independent variable of our key interest is the ownership type of the venture. Specifically, *Foreign VC Participation* is equal to 1 if at least one foreign VC invests in a financing round and 0 otherwise. Other independent variables that may have an impact on venture exit performance are also included, for instance, dummies indicating the stage of investment that includes seed, early and expansion; the type of industry with computer, communication, semiconductor, biotechnology and medical/healthcare as the choices; the lead VC characteristics with independent VC, corporate VC, VC age, VC's total investment experience, VC's investment experience in the entrepreneur's country, measures of geographic distance and cultural distance between VCs and entrepreneurs. The geographic distance between VC and the venture is measured as the distance between the capitals (or the most populated cities if the capital is sparsely populated) of the respective countries of the VC and the venture.¹² We use Hofstede's (2001) measures of the four dimensions of a society's culture, and then use the differences in the measures to capture the idea of "cultural

¹² We obtain the distance data from the CEPII website. Please see <http://www.cepii.fr/anglaisgraph/bdd/distances.htm>.

distance” between countries.¹³ Hofstede classifies culture into four major dimensions — small versus large power distance, uncertainty avoidance, individualism versus collectivism, and masculinity versus femininity. *Power distance* measures the degree of equality, or inequality, between people in the country's society. *Uncertainty avoidance* captures the society's attitude towards uncertainty and its attempt to cope with anxiety in uncertain situations. *Individualism* refers to the extent the society helps reinforcing the individual achievement, whereas *collectivism* emphasizes collective action by individuals. *Masculinity* reflects the extent to which the society values the traditional masculine features such as assertiveness, achievement, competitiveness, and the accumulation of materialistic possessions. In contrast, *femininity* emphasizes relationships and quality of life. In all specifications, we further control for the size of investment, country legality and stock market development. Year and entrepreneurial firm country fixed effects are also used.

Hazard ratios instead of coefficients are reported. A hazard ratio of an independent variable that is smaller (greater) than 1 indicates a lower (higher) probability of exit than the comparison group. We find that ventures with foreign VC participation at the round level are significantly more likely to exit through either IPO or acquisitions, nevertheless, not necessarily with more success if measured by the probability of IPO alone. Among other independent variables, we show that seed stage and early stage ventures are significantly less likely to exit successfully, while ventures with large investment size have higher probability of successful exits, consistent with the existing literature. Further, we show that at the round level, the cultural distance has a significantly negative effect on the exit performance, supporting *Hypothesis 2b*.

In the second set of regressions (the last three models), we apply probit regressions to analyze the firm level exit performance. Similar to the round level analysis, we examine the probability of IPOs, the probability of IPOs and acquisitions, and the probability of IPOs and acquisitions with valuation data. The independent variable of our key interest is the firm level ownership type. *Foreign VC participation* is equal to 1 if at least one foreign VC invested in the venture in any of the rounds, and 0 otherwise. Our other independent variables are similar to those used in the round level analysis except that the lead VC is defined as the one that invests the largest amount in the very first round and stage dummies indicate the stage of the venture when they receive the very first round.

At the firm level, we do not find evidence that ventures with foreign VC participation are more likely to exit successfully. Among other independent variables, we show that ventures that receive VC funding at their seed or early stages are less likely to exit successfully, while ventures with larger investment amounts have significantly higher probability to exit either through IPO or successful acquisitions. These findings are consistent with those from the round level analysis.

5.2.2. Joint investments by foreign and local VCs vs. foreign VCs alone

In this section, we analyze the choice of joint investment by foreign and local VCs and foreign VCs alone and its implication for the exit performance. Panel A of *Table 5* reports the results of probit regressions on the choice of joint investments by foreign and local VCs and foreign VCs alone at the round level and the firm level, respectively. Marginal effects instead of coefficients are reported. In the first regression, the dependent variable is an indicator variable which is equal to 1 if at least one foreign VC and one local VC invested in a specific round; and 0 otherwise. The independent variables include measures of information asymmetry: (a) round size and number; (b) the stage of ventures at the financing round date; (c) the industries in which the ventures reside. In addition, we control for lead VC characteristics such as independent VC dummy, Corporate VC dummy, whether the VC fund is a seed or early stage fund, VC age, VC's total investment experience, VC's experience in the entrepreneur's country, the geographic distance and cultural distance between the VC and the venture. The dependent variable in the second regression is an indicator variable which is equal to 1 if at least one foreign VC and one local VC participated in any of the ventures' financing rounds; and 0 otherwise. Independent variables are similar to those used in the round level analysis except that the lead VC is defined as the one that invests the largest amount in the very first round and stage dummies indicate the stage of the venture when they receive the very first round. In both specifications, we also control for country legality, stock market development, year fixed effect and entrepreneur's country fixed effect.

We show that by forming partnership with local VCs, both at the round level and the firm level, foreign VCs invest more aggressively in early stage ventures. In terms of economic significance, foreign VCs are about 10–23% (12–16%) more likely to invest in the seed (early) stage ventures when investing together with local VCs than when investing alone. We further show that foreign VCs are 7–12% more likely to invest in computer-related, 13–16% more likely to invest in semiconductor, and over 30% more likely to invest in biotechnology ventures if they partner with local VCs than if they are investing alone. This finding supports the idea of *Hypothesis 3a* where working with local VCs provides a practical solution to the information asymmetry problem.

In addition, we find that foreign VCs with more investment experience in the entrepreneur's country are more likely to form partnerships. Specifically, one standard deviation increase in the $\ln(\text{Investment experience in entrepreneur's country})$ increases the probability of joint investment by 5% at the round level and 18% at the firm level. Both geographic distance and cultural distance between the countries of VCs and ventures significantly decrease the probability of forming partnership between foreign and local VCs, supporting *Hypothesis 2a*. It is well documented that countries with higher cultural distance display higher mutual distrust (Chakrabarti et al., 2008; Guiso et al., 2008). Our finding indicates the possibility of distrust between foreign VCs and local VCs due to cultural differences could be a serious hurdle for them to form partnerships. Interestingly, we show that country legality index is significantly and negatively correlated with the probability of joint investment.

In Panel B of *Table 5*, we compare the exit performance of ventures that are jointly invested by foreign and local VCs and those invested by foreign VCs alone. Similar to Panel B of *Table 4*, we analyze successful exits in three scenarios, including IPOs only, IPOs and acquisitions, and IPOs and acquisitions with valuation data. The first three regressions are the round level analysis. In the first regression, we apply the competing risk model, where the failure event is IPO exit and the competing event is unsuccessful exit. Those

¹³ Specifically, we compute the cultural distance as follows: $\text{Hofstede cultural distance} = \left(\frac{\sum_{i=1}^4 (H_{VCi} - H_{EFi})^2}{4} \right)^{1/2}$, where H_{VCi} is the Hofstede measure of the lead VC's origin country on cultural dimension i , and H_{EFi} is the Hofstede measure of the entrepreneurial firm's country on cultural dimension i .

Table 5

Joint investments by foreign and domestic VCs vs. investments by foreign VCs only.

In Panel A, we use probit regressions to investigate whether there is systematic difference in investment criteria across VC ownership types (joint investments by foreign and local VCs vs. investments with foreign VCs alone) at the round level and the firm level, respectively. The dependent variable in the first (second) regression is a dummy which is set to equal to one if at least one foreign VC and one local VC invest in a specific financing round (in any of the financing rounds) and 0 otherwise. All regressions include entrepreneurial firm country fixed effect and year fixed effect. The independent variables include measures of information asymmetry: (a) round size and number; (b) the stage of ventures at the financing round date; (c) the industries in which the ventures reside. In addition, we control for lead VC characteristics such as independent VC, Corporate VC, whether the VC fund is a seed or early stage fund, VC age, total investment experience, VC's investment experience in a specific Asian country, the geographic and cultural distance between lead VC and the venture. Definitions of other independent variables are provided in the Appendix. The independent variables in the second regression are similar to those used in the round level analysis except that the lead VC is defined as the one that invests the largest amount in the very first round and stage dummies indicate the stage of the venture when they receive the very first round. Marginal effects instead of coefficients are reported. In Panel B, we analyze the impact of VC ownership type (joint investments by foreign and local VCs vs. investments with foreign VCs alone) on the likelihood of successful exits. For successful exits, we consider three scenarios: IPOs only, IPOs and M&As, and IPOs and M&As with valuation data. The first three regressions represent the round level analyses. In the first regression, we apply the competing risk model, where the failure event is IPO exit and the competing event is unsuccessful exit. Those that have not exited are treated as right censored. In the second and third regressions, we utilize Cox Proportional Hazard model. The dependent variable of these three regressions is the natural logarithm of time to an exit which is measured from the date of each round. Hazard ratios instead of coefficients are reported. The independent variable of our key interest is the ownership type of the venture. Specifically, *joint investment* is equal to 1 if at least one foreign VC and one local VC invest in a financing round and 0 otherwise. Other independent variables that may have an impact on venture exit performance are also included, for instance, dummies indicating the stage of investment that includes seed, early and expansion; the type of industry with computer, communication, semiconductor, biotechnology and medical/healthcare as the choices; the lead VC characteristics with independent VC, corporate VC, VC age, VC's total investment experience, VC's investment experience in the entrepreneur's country, measures of geographic distance and cultural distance between VCs and entrepreneurs. The last three regressions are the firm level analyses. We utilize probit regressions in all three specifications, where the dependent variable is a dummy which is equal to 1 if the venture has exited as an IPO (IPO or Acquisition, IPO or Acquisition with valuation data) by the end of 2007, and 0 otherwise. At the firm level analyses, *joint investment* is equal to 1 if at least one foreign VC and one local VC invested in the venture in any of the rounds, and 0 otherwise. The independent variables are similar to those used in the round level analysis except that the lead VC is defined as the one that invests the largest amount in the very first round and stage dummies indicate the stage of the venture when they receive the very first round. All regressions include year fixed effect and entrepreneurial firm nation fixed effect. ***, **, and * represent significance at the 1%, 5%, and 10% confidence levels, respectively.

Panel A: selection		
	Round level	Firm level
Round		
Round size	0.041*** (0.000)	
Round 1	−0.013 (0.610)	
Stage		
Startup/seed	0.103* (0.074)	0.231*** (0.002)
Early stage	0.117*** (0.008)	0.161** (0.041)
Expansion stage	0.022 (0.522)	0.022 (0.768)
Industry		
Computer related	0.076** (0.023)	0.116* (0.052)
Communication and media	0.044 (0.232)	0.094 (0.144)
Semiconductor	0.129*** (0.007)	0.160* (0.052)
Biotechnology	0.367*** (0.007)	0.176 (0.176)
Medical/health care	0.078 (0.275)	0.145 (0.225)
Lead VC characteristics		
Independent VC	−0.043 (0.234)	−0.108** (0.045)
CVC	0.090* (0.089)	0.134* (0.076)
Seed or early stage fund	0.044 (0.271)	−0.033 (0.601)
Ln(VC firm age)	−0.003 (0.882)	0.078** (0.010)
Ln (total investment experience)	0.051*** (0.000)	0.013 (0.236)
Ln (investment experience in entrepreneur's country)	0.032** (0.044)	0.128*** (0.000)
Ln (geographic distance)	−0.080*** (0.000)	−0.168*** (0.000)
Cultural distance	−0.006** (0.047)	−0.016*** (0.007)
Other control variables		
Country legality index	−0.052*** (0.000)	−0.049*** (0.001)

Table 5 (continued)

Panel A: selection									
	Round level							Firm level	
Stock market development	−0.135** (0.044)							0.061 (0.564)	
Year fixed effect	Yes							Yes	
Entrepreneurial firm nation fixed effect	Yes							Yes	
N	1619							754	
Pseudo R ²	23.85%							40.49%	
Panel B: exit performance									
	Round level			Firm level					
	IPO	IPO and ACQ	IPO and ACQ (with valuation)	IPO		IPO and ACQ		IPO and ACQ (with valuation)	
Ownership type									
Joint	1.037 (0.877)	1.157 (0.417)	1.145 (0.461)	0.046** (0.033)	0.057** (0.026)	0.064** (0.041)	0.090** (0.046)	0.072*** (0.005)	0.069* (0.052)
Stage									
Startup/seed	0.430 (0.106)	0.687 (0.315)	0.643 (0.264)	−0.010 (0.782)	−0.021 (0.414)	0.071 (0.303)	0.061 (0.365)	0.035 (0.517)	0.025 (0.627)
Early stage	0.734 (0.345)	0.741 (0.264)	0.722 (0.244)	0.006 (0.857)	−0.005 (0.855)	0.084 (0.154)	0.072 (0.207)	0.013 (0.759)	0.005 (0.905)
Expansion stage	1.421 (0.165)	1.179 (0.460)	1.221 (0.372)	0.025 (0.421)	0.012 (0.645)	0.084* (0.089)	0.077 (0.109)	0.048 (0.218)	0.041 (0.271)
Industry									
Computer related	1.063 (0.796)	1.004 (0.983)	0.940 (0.754)	0.018 (0.507)	0.007 (0.765)	0.038 (0.356)	0.030 (0.455)	0.026 (0.431)	0.018 (0.568)
Communication and media	0.570* (0.062)	0.840 (0.460)	0.789 (0.323)	−0.020 (0.443)	−0.027 (0.172)	0.008 (0.846)	0.001 (0.985)	−0.007 (0.865)	−0.014 (0.639)
Semiconductor	0.622 (0.162)	0.824 (0.489)	0.695 (0.217)	0.033 (0.449)	0.017 (0.637)	0.068 (0.282)	0.054 (0.378)	0.017 (0.721)	0.003 (0.950)
Biotechnology	1.340 (0.606)	0.754 (0.588)	0.891 (0.833)	0.006 (0.911)	−0.017 (0.639)	0.020 (0.830)	−0.002 (0.984)	−0.007 (0.919)	−0.025 (0.651)
Medical/health care	0.754 (0.626)	0.840 (0.739)	0.813 (0.693)	0.049 (0.450)	0.035 (0.527)	0.075 (0.401)	0.063 (0.463)	0.061 (0.423)	0.046 (0.517)
Lead VC characteristics									
Independent VC	0.674* (0.079)	0.649** (0.038)	0.643** (0.038)	−0.011 (0.641)	−0.007 (0.743)	0.014 (0.682)	0.016 (0.630)	−0.001 (0.973)	−0.001 (0.987)
CVC	0.969 (0.926)	0.937 (0.819)	0.925 (0.788)	−0.017 (0.846)	−0.013 (0.600)	−0.025 (0.575)	−0.026 (0.558)	−0.011 (0.771)	−0.013 (0.716)
Ln(VC firm age)	0.854 (0.240)	0.917 (0.454)	0.904 (0.392)	0.002 (0.846)	0.005 (0.632)	0.001 (0.976)	0.004 (0.836)	0.013 (0.391)	0.016 (0.297)
Ln(total investment experience)	1.131* (0.093)	1.043 (0.517)	1.103 (0.139)	0.002 (0.620)	0.001 (0.945)	−0.003 (0.653)	−0.005 (0.445)	−0.002 (0.723)	−0.004 (0.452)
Ln(investment experience in entrepreneur's country)	0.923 (0.446)	1.008 (0.941)	0.951 (0.640)	−0.002 (0.867)	0.003 (0.744)	−0.008 (0.586)	−0.003 (0.844)	−0.004 (0.734)	−0.001 (0.924)
Ln(geographic distance)	1.503*** (0.004)	1.207* (0.077)	1.219* (0.068)	0.018 (0.153)	0.023** (0.049)	0.015 (0.315)	0.017 (0.253)	0.006 (0.626)	0.008 (0.517)
Cultural distance	0.925*** (0.001)	0.953*** (0.003)	0.946*** (0.003)	−0.004 (0.154)	−0.004* (0.059)	−0.001 (0.930)	−0.001 (0.843)	−0.001 (0.667)	−0.001 (0.603)
Other control variables									
Country legality index	0.917 (0.284)	0.990 (0.877)	0.982 (0.781)	−0.011 (0.132)	−0.011* (0.087)	0.013 (0.133)	0.013 (0.149)	−0.004 (0.586)	−0.004 (0.556)
Stock market development	0.669 (0.400)	0.549 (0.150)	0.490* (0.096)	0.038 (0.336)	0.032 (0.351)	−0.015 (0.806)	−0.020 (0.742)	−0.006 (0.899)	−0.013 (0.785)
Ln(total investment)	1.253*** (0.001)	1.219*** (0.002)	1.278*** (0.000)	0.032*** (0.000)	0.018** (0.010)	0.060*** (0.000)	0.050*** (0.000)	0.046*** (0.000)	0.035*** (0.000)
Syndicate of 2 investors					0.054 (0.175)		0.013 (0.799)		0.029 (0.526)
Interaction: joint * syndicate of 2					−0.081*** (0.000)		−0.107*** (0.007)		−0.066** (0.039)
Syndicate of 3 or more investors					0.055 (0.104)		0.039 (0.432)		0.058 (0.173)
Interaction: joint * syndicate of 3 or more					−0.038** (0.040)		−0.039 (0.311)		−0.020 (0.510)

(continued on next page)

Table 5 (continued)

Panel B: exit performance									
	Round level			Firm level					
	IPO	IPO and ACQ	IPO and ACQ (with valuation)	IPO		IPO and ACQ		IPO and ACQ (with valuation)	
Other control variables									
Year Fixed Effect	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Entrepreneurial firm nation fixed effect	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
N	1619	1619	1619	754	754	754	754	754	754
Pseudo R ²				14.37%	19.00%	10.84%	12.31%	12.79%	14.44%
Log pseudolikelihood	−959.68	−1156.31	−1101.52						
Interaction: joint * syndicate of 2					−0.081*** (0.000)		−0.107*** (0.007)		−0.066** (0.039)

that have not exited are treated as right censored. The dependent variable of the competing risk regression is the natural logarithm of time to an exit which is measured from the date of each round. We utilize the Cox Proportional Hazard model in the second and third regressions. The independent variable of our key interest is the ownership type of the venture. Specifically, *joint investment* is equal to 1 if at least one foreign VC and at least one local VC invested in a financing round and 0 otherwise. In the remaining six regressions, we perform probit regressions on the probability of successful exits at the firm level. The key variable of our interest is *joint investment* at the firm level which is equal to 1 if at least one foreign VC and at least one local VC invested in the venture (not necessarily the same round) and 0 otherwise. Other independent variables are similar to the ones we use in Panel B of Table 4.

At the round level, we do not find that joint investments outperform ventures invested by foreign VCs alone. We find that cultural distance is significantly and negatively associated with the exit performance, further supporting *Hypothesis 2b*. But, we find that geographic distance is positively associated with exit performance. This result is seemingly in contrast with [Chemmanur et al. \(2010\)](#). This is explained by the fact that U.S. or European VCs, in our study, are associated with longer geographical distances (the distance between the capital cities of two nations where the lead VC and venture are headquartered) than Asian VCs, while these VCs are likely to be associated with better exit performance due to their broader experience, skills, and networking.

At the firm level, when we do not control for the syndicate size, we find that joint investments are significantly associated with higher probability of successful exits, no matter which measure is used. Specifically, other things equal, joint investments are 5% more likely to exit through IPOs, and 6–7% more likely to exit either through IPOs or acquisitions, supporting *Hypothesis 3b*.

As shown in Table 3, joint investments often have a greater number of investors at the firm level. Many existing studies have shown that syndicate size has a positive impact on the venture's performance. To further differentiate whether the better exit performance of joint investments is due to the syndicate size effect or the effect of the unique foreign–local partnership, we add controls for the syndicate size and the interaction terms between joint ownership type and syndicate size. Specifically, we categorize ventures into three groups, with one investor, two investors, and three or more investors. The base group is ventures with only one investor (foreign VC alone). We use the interaction between joint ownership type with syndicate of two investors and syndicate of three or more investors, respectively. With these additional controls for the syndicate size, we continue to find that joint investments have significantly higher probability (by 6–9%) for successful exits than those that are invested by foreign VCs alone.

5.2.3. Robustness check

An additional issue that may affect our findings is that the ownership type of the local ventures in these emerging economies is not randomly determined. It is well established in the VC literature that VCs with more experience and broader networks are more likely to attract high-quality entrepreneurial firms and provide high-quality value-adding services to entrepreneurial firms ([Gompers et al., 2010](#); [Hellmann and Puri, 2002](#); [Hsu, 2004](#)). As we discuss and show earlier, foreign VCs have relative advantages over local VCs (in countries with less developed VC industry) in that they are often more experienced, more professionalized, and larger in size. These factors may allow foreign VCs to attract the most talented entrepreneurs and the best quality ventures, which have important implications for the ultimate success of the ventures. In other words, the better performance of ventures invested jointly by foreign and local VCs thus could be driven by the fact that they are of the best quality at the very beginning.¹⁴ To make sure our findings are not contaminated by this potential endogenous selection bias, we apply the propensity score matching methodology to our data.

The method of propensity score matching has been receiving more attention in social studies mainly for dealing with the endogeneity issues since the seminal work of [Rosenbaum and Rubin \(1983\)](#). There are several steps involved. The first step is to estimate the conditional probability of a specific venture ownership type given the observable characteristics with a probit regression. The independent variables we include in the probit regression are dummies indicating the stage and industry of the venture, various characteristics of the lead VC and country level control variables (similar to the ones used in our selection models at the firm level). The second step is to estimate the conditional probability or the propensity score for each observation from the probit model. As a third step, we use various matching techniques, such as the nearest neighbor matching, radius matching, and

¹⁴ Our literature search suggests that there is no universally accepted direct measure of venture quality. Some use the number of patents ([Hsu and Ziedonis; Kortum and Lerner, 2000](#)). Unfortunately, such data is not available for the six Asian countries we examine. We use several indirect measures of venture quality such as expansion stage, round size or total investment size, investment experience of the VCs and the size of the syndicate as second-best proxies. These measures are commonly used in the existing literature as indication of better venture quality ([Gompers, et al., 2010](#)).

Table 6

Robustness check: propensity score matching.

In this table, we conduct the propensity score matching analysis on the differences in the probability of IPO and the probability of IPO and M&As with valuation data conditional on ownership types, which controls for the endogeneity of companies receiving foreign VC investments. We report the results using three different matching methods, including nearest neighbor matching, radius matching, and kernel matching. The t-statistics based on bootstrapped standard errors are reported in parentheses. ***, **, and * represent significance at the 1%, 5%, and 10% confidence levels, respectively.

Propensity score matching methods	Difference in the probability of IPO			Difference in the probability of IPO and ACQ (with valuation data)		
	Foreign alone vs. domestic alone	Joint vs. domestic alone	Joint vs. foreign alone	Foreign alone vs. domestic alone	Joint vs. domestic alone	Joint vs. foreign alone
Nearest neighbor matching	0.025 (1.079)	0.052*** (2.874)	0.055* (1.730)	0.041 (1.604)	0.078*** (3.760)	0.073 (1.598)
Radius matching	0.039** (2.287)	0.050*** (3.109)	0.052** (2.078)	0.054*** (2.971)	0.079*** (4.468)	0.084*** (2.894)
Kernel matching	0.019 (0.899)	0.049*** (3.082)	0.053* (1.830)	0.033 (1.513)	0.077*** (3.798)	0.105** (2.745)

the Kernel matching to form sub-groups of observations with different ownership type but with equal or close propensity scores.¹⁵ As a final step, we compute the likelihood of IPOs and the probability of IPOs or successful acquisitions for these sub-groups. The differences in the likelihood of successful exits then reflect the contribution to the success by different ownership type after controlling for the endogeneity of selection.

As shown in Table 6, ventures jointly invested by foreign and local VCs consistently outperform those invested by local VCs alone and those invested by foreign VCs alone no matter which matching technique we apply. Specifically, ventures jointly invested by foreign and local VCs are about 5–10% more likely to successfully exit than those invested by local VCs or those invested by foreign VCs alone. The findings of propensity score matching further underline the results from our multivariate regression analysis in Panel B of Table 5 that partnership between foreign and local VCs is an effective approach to deal with the information friction associated with both geographic and cultural distances when foreign VCs explore their investment opportunities in the international venture capital market.

6. Conclusions

In this paper, we study the investment behavior of VCs while they pursue expansion outside their home locations, specifically, VCs expanding their investments to the emerging Asian markets. Several interesting findings emerge from our studies. We believe these findings not only fill some of the voids in the existing literature on international venture capital investments, but also have helpful implications for foreign VCs and local entrepreneurs. We discuss our major findings, their relation to the existing literature, and their implications for academics and practitioners in details below.

First, we find that foreign VCs play a noteworthy role in the Asian venture capital industry. During the sample period we study (1996–2006), foreign venture capital accounts for about 75% of the funding infused to the local entrepreneurial firms according to VentureXpert. Second, we examine how the frictions associated with cultural disparity and geographical distance impact VCs' behavior when they explore their investment opportunities in unfamiliar territory. We find that foreign VCs are less likely to invest in seed or early stage firms or participate in the very first round, which is consistent with these VCs avoiding more informationally opaque investments. Forming partnership with local VCs in the local areas effectively helps reduce such frictions. The cultural distance, however, gets sometimes in the way of forming partnership between foreign and local VCs. Our paper shows that information asymmetry and cultural disparity not only matters for foreign VCs to select local entrepreneurs, but also matters for successful exit performance. Third, as predicted, we find that the entrepreneurial firms invested jointly by foreign and local VCs are more likely to successfully exit through IPOs and acquisitions. Our results remain robust even after we control for the endogeneity of the selection between VCs and entrepreneurs using propensity score matching.

One of the concerns, though, is the completeness and reliability of the dataset we use for this study. VentureXpert is one of a few databases that collect venture capital data in the Asian market. It is acknowledged that the database is quite comprehensive on covering the U.S. venture capital investments but has less developed coverage in Europe and Asia. Further, the foreign investments by U.S. VCs are likely better covered than those by VCs located in other economies. This introduces two potential biases in the data. First, U.S. VCs are over-represented in our sample. To make sure our results are not driven by U.S. VCs, in unreported analysis, we exclude all U.S. VCs, and redo the analysis for the non-U.S. foreign VCs. Our major findings hold with such a sub-sample. Second, entrepreneurial firms invested by small local VCs alone are likely to be missing in our sample. To improve the quality of our sample data, we focus our analysis on the relatively recent period (1996–2006) when foreign investments in the Asian markets we study are more prevalent and the data collected by VentureXpert are of relatively higher quality, although still biased.¹⁶

¹⁵ Details for the various matching methods are provided in STATA 11 manual.

¹⁶ When investigating a period from 1980 to 2006, the empirical results are qualitatively similar.

As the amelioration of cross-border VC investment into the Asian emerging economies continues, we expect that future studies that examine the evolution of the foreign VC and local VC and entrepreneurs coalition over time would significantly contribute to our understanding of global cross-border VC investment selection and performance. Foreign VCs may help professionalize the local entrepreneurial firms given their experience of advising and nurturing ventures in their home countries. These local entrepreneurial firms may further have spillover effects on their fellow firms which are currently not financed by foreign venture capital. The presence of foreign venture capital firms and their partnership with local VCs may help professionalize the local venture capital firms. This potentially includes, but is not limited to, adopting standard venture capital contracts, better knowledge on advising and nurturing entrepreneurs, and better knowledge on constructing deals and taking those firms to initial public offerings or other exits on both the domestic markets and international markets. These topics invite further in-depth studies in the future. We conjecture that hiring local talents would provide a practical solution to this issue as found in other studies on service companies pursuing their international expansion. It is an interesting research question regarding how mechanisms, such as expatriating staff and hiring local talents, can effectively overcome hurdles related to information friction and cultural differences in international venture capital investments. We believe this also remains an important, albeit unexplored, area in international venture capital investments.

Appendix A. Definition of variables

Variable name	Definition
Startup/seed ^a	A dummy which is equal to 1 if the venture is at the startup/seed stage, and 0 otherwise
Early stage	A dummy which is equal to 1 if the venture is at the early stage, and 0 otherwise
Expansion stage	A dummy which is equal to 1 if the venture is at the expansion stage, and 0 otherwise
Computer related	A dummy which is equal to 1 if the venture is in the computer-related industry, and 0 otherwise
Communication/media	A dummy which is equal to 1 if the venture is in the communication/media industry, and 0 otherwise
Semiconductor	A dummy which is equal to 1 if the venture is in the semiconductor industry, and 0 otherwise
Biotechnology	A dummy which is equal to 1 if the venture is in the biotechnology industry, and 0 otherwise
Medical/health care	A dummy which is equal to 1 if the venture is in the medical/health care industry, and 0 otherwise
VC firm age	The difference between investment date and VC firm founding date, measured in years
N of companies VC invested	Number of entrepreneurial firms that a VC firm had invested before
Round size	The natural logarithm of total amount of capital raised in a specific round
Round 1	A dummy which is set to equal to 1 if it is the first round for a specific entrepreneurial firm, 0 otherwise
Independent VC	A dummy which is set to equal to 1 if the VC is an independent professional VC firm, 0 otherwise
CVC	A dummy which is set to equal to 1 if the VC is corporate venture capital firm, 0 otherwise
Seed or early stage fund	A dummy which is set to equal to 1 if the VC fund specializes in investing in seed or early stage ventures, 0 otherwise
Ln(VC firm age)	The natural logarithm of VC firm age
Ln(total investment experience)	The natural logarithm of the total number of entrepreneurial firms that a VC firm had invested before
Ln(experience in entrepreneur's country)	The natural logarithm of the total number of entrepreneurial firms that a foreign VC firm had invested in one of the six Asian countries before
Ln(geographic distance)	The natural logarithm of the geographic distance between the capital cities of the two nations where the lead VC and the venture are headquartered
Cultural distance	Hofstede cultural distance = $\frac{(\sum_{i=1}^4 (H_{VCi} - H_{EFi})^2)^{1/2}}{4}$, where H_{VCi} is the Hofstede measure of the lead VC's origin country on cultural dimension i , and H_{EFi} is the Hofstede measure of the entrepreneurial firm's country on cultural dimension i .
Country legality index	The value of legality is the weighted average following factors (based on Berkowitz et al., 2003, and Cumming et al., 2006): civil versus common law systems, efficiency of judicial system, rule of law, corruption, risk of expropriation, risk of contract repudiation, shareholder rights (as in La Porta et al., 1997, 1998). Higher value of legality index indicates "better" legal system.
Stock market development	The average of the three normalized indices, including Stock market capitalization/GDP, Total value of shares traded/GDP, and Total value of shares traded/stock market capitalization.
Foreign VC participation: round level	A dummy which is set to equal to 1 if at least one foreign VC invested in a specific round, 0 otherwise
Joint investment: round level	A dummy which is set to equal to 1 if at least one foreign VC and at least one local VC invested in a specific round, 0 if there is no local VC participation
Foreign VC participation: firm level	A dummy which is set to equal to 1 if at least one foreign VC invested in an entrepreneurial firm, 0 otherwise
Joint investment: firm level	A dummy which is set to equal to 1 if at least one foreign VC and at least one local VC invested in an entrepreneurial firm, 0 if there is no local VC participation
Ln(total amount invested)	At the round level analysis, total amount invested refers to the round size; at the firm level analysis, total amount invested refers to the total amount invested from the very first round until exit
Syndicate of 2 investors	A dummy which is equal to 1 if there are two investors invested in the entrepreneurial firm, and 0 otherwise
Syndicate of 3 or more investors	A dummy which is equal to 1 if there are three or more investors invested in the entrepreneurial firm, and 0 otherwise

^aThe stage dummies (Seed, Early, Later) are based on VentureXpert definition. Seed stage is defined as an investment strategy involving portfolio companies that have not yet fully established commercial operations, and may also involve continued research and product development. Early stage is defined as an investment strategy involving investments in companies for product development and initial marketing, manufacturing and sales activities. Later stage describes funds that make investments into portfolio companies that have an already established product or service that has already generated revenue, but may not be making a profit. Later stage funds make the last round of investments in portfolio companies before an exit in the form of an IPO by a strategic partner.

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